



A Survey of Multi-AI Agent Collaboration: Theories, Technologies and Applications

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Abstract

As an important application of large language model (LLM), artificial intelligence agent (AI Agent) have the ability to autonomously perceive, understand, plan, memory, act, and use tools. It can automate complex tasks and effectively empower various business scenarios. Single AI Agent flexible and diverse deployment, multi-AI Agent innovative interaction and collaboration, multi-AI Agent collaboration improves the autonomy of the intelligent system by integrating the capabilities of single AI Agent. This paper provides an overview of multi-AI Agent from four aspects. Firstly, the core capabilities of AI Agent were outlined, and multi-AI Agent collaboration was introduced and its characteristics were analyzed. Secondly, the theoretical basis, key technologies, and scenario applications of multi-AI Agent collaboration were discussed, and the mechanism, architecture design, communication protocol, reinforcement learning, security and trustworthiness of multi-AI Agent collaboration were deeply studied. Thirdly, the advantages and disadvantages of multi-AI Agent collaboration in technology, application, and security directions were summarized, and frontier research and innovation directions were provided. Finally, a summary and outlook were made on the high-quality development of multi-AI Agent collaboration.

CCS Concepts

• **General and reference** → Document types; Surveys and overviews, Computing methodologies; Artificial intelligence; • **Software and its engineering** → Software creation and management; Software development techniques.

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Keywords

Large language model, Multi-AI Agent collaboration, Collaborative mechanism, Communication protocol, Reinforcement learning

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1 Introduction

In recent years, the technological capabilities of LLM have continuously improved, laying a solid technical foundation for scenario applications and user interaction. As an important application mode of LLM, AI Agent transform model capabilities into practical problem-solving abilities, bridging the gap between business scenarios and human interaction. The single AI Agent independently completes all reasoning, planning and tool execution through a single model driven. However, in practical applications, it faces multidimensional constraints such as insufficient efficiency, limited understanding, low fault tolerance, poor observability, and single task requirements, making it difficult to meet complex task requirements.

With the continuous emergence of collaborative demands across scenarios, domains, and levels, AI Agent technology is gradually being iteratively upgraded towards multimodal interaction and multi-AI Agent collaboration. Multi-AI Agent drives technological and intelligent advancements through the complementarity of cooperation and competition. Multi-AI Agent collaboration [1] integrates and optimizes the capabilities of the single AI Agent, consisting of two or more AI Agents. Different AI Agents can use the same or different models, access the same or different tools, and play different roles, effectively solving the problem of a single ability being unable to meet complex task requirements. Multi-AI Agent collaborative systems can effectively reduce the difficulty of executing complex tasks, improve task execution efficiency, and further enhance the intelligence level of AI Agents through dynamic task decomposition, distributed resource scheduling, and reasonable task allocation capabilities.

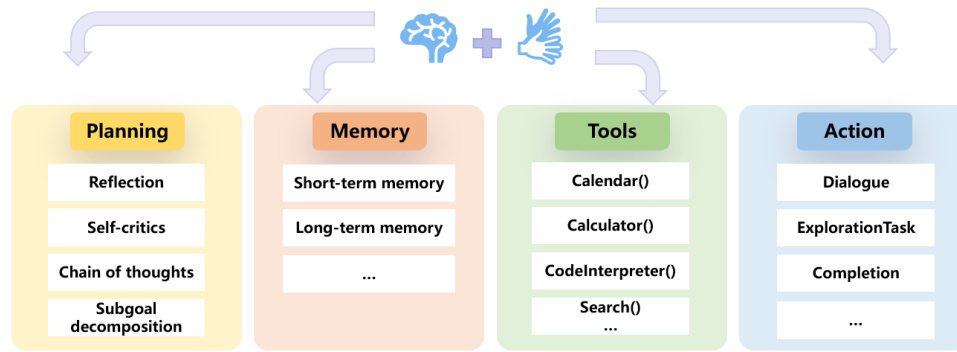


Figure 1: Core capabilities of AI Agent.

Based on the current development status of multi-AI Agent systems, this paper will first introduce the core capabilities and key technologies of AI Agent, and focus on the theoretical basis, key technologies, and scenario applications of multi-AI Agent collaboration, including collaboration mechanisms, architecture design, communication protocols, and other theoretical foundations. It will cover key technologies such as collaboration, learning, and security, as well as the application penetration status of multi-AI Agent collaboration in various industries. In depth research will be conducted to explore the development advantages and disadvantages of multi-AI Agent collaboration in detail, and innovative research directions will be sorted out. Finally, the technical application prospects and future development of multi-AI Agent collaboration will be discussed, which can provide reference for the study of key issues related to multi-AI Agent collaboration.

2 Research Fundamentals

2.1 Core capabilities of AI Agent

AI Agent can be used as a typical form of native application of AI with good results in planning and decision-making, tool calling, long-term memory and task execution. The specific information is shown in Figure 1.

AI Agent's planning technique has great advantages. In complex environments, it can develop optimal action strategies to achieve specific goals. By combining LLMs, real business systems and task scheduling mechanisms, AI Agent has significant capabilities in multi-level planning and task optimization. SELA [2] combines monte carlo tree search (MCTS) with LLM based AI Agent for automatic optimization in machine learning tasks. Aflow [3] implements workflow optimization using MCTS to dynamically explore the optimal agents combination and execution process in multi-tasking, greatly improving the processing efficiency of complex tasks.

The AI Agent's memory mechanism [4] simulates the human memory process and consists of two parts: short-term memory, which contains only context, and long-term memory, which can be permanent. The long-term memory accumulates the AI Agent's historical experience and plays a central supporting role in its logical reasoning and planning.

In terms of tooling techniques, it enables AI Agents to interact with the external environment in real time, enabling them to perform complex tasks and decisions. By using function call techniques, AI Agents can connect to external systems using structured inputs, such as interacting through APIs for autonomous decision making and reliable parsing. SPO [5], an open source self-supervised prompt optimization tool, allows AI to autonomously optimize prompt words to make them suitable for both closed-ended tasks and open-ended creative scenarios.

By invoking external tools, applying internal knowledge and interacting with the environment, the AI Agent's decisions are translated into concrete actions and generate positive feedback to the environment. Control algorithms, path planning algorithms and inverse kinematics solver algorithms together form the basis of the actions taken by the AI Agent to perform various specific tasks and achieve pre-defined goals.

2.2 Key technologies of AI Agent

Based on LLM, AI Agent can bridge the interaction gap between LLM and users, transform modelling capabilities into actual problem solving capabilities with sensing, judging, responding and learning capabilities [6], bridging the last mile of LLM implementation.

The AI Agent Building Framework greatly simplifies the agent development process through its pre-built components, standardized architecture and convenient tools. The framework's customization and integration capabilities [7] are so powerful that it can effectively address key challenges such as task orchestration, multimodal integration and scalability, greatly expanding the application scenarios and possibilities. Supported by scientific task planning, multiple iterations and optimization techniques, the AI Agent workflow [8] can help produce higher quality results, improve the controllability of task decomposition and process planning, significantly improve model performance, and ensure smooth and efficient task completion. The problem of deploying and running AI Agents on devices with limited computing power, memory and energy consumption can be solved by end-to-end intelligence techniques. For example, model compression and hardware acceleration can be combined to balance performance and efficiency, reduce cloud dependency, and improve responsiveness and privacy. In terms of multi-AI Agent

collaboration [9], resource allocation is optimized by rationally decomposing tasks, sharing knowledge, executing subtasks and adjusting goals, which significantly improves the ability to perform complex tasks and solve problems, and helps AI Agents achieve quality and efficiency.

2.3 Characteristics of multi-AI Agent collaboration

The rapid development of AI Agent technology has paved the way for multi-AI Agent collaboration. Multi-AI Agent collaboration system (MACS) is a system consisting of multiple autonomous and independent AI Agents. Each AI Agent is capable of sensing the environment, making decisions and performing tasks. At the same time, they can share information, coordinate tasks and take collaborative actions to achieve the overall goal, characterised by specialization, autonomy, robustness, interpretability and scalability [10].

Each AI Agent focuses on a specific domain or task and has specialized knowledge, skills and experience in that domain, allowing it to make more accurate and professional decisions and recommendations. At the same time, the AI Agent can make decisions and take actions independently, based on its own perceptions and goals, without the need for external guidance or intervention. The collaborative operation of AI Agents within the system is based on real-time feedback to adjust task strategies. In terms of robustness, multi-AI Agent systems coordinate their actions and complete tasks together through local information and mutual communication. This distributed nature also makes the system more flexible and robust, and when a single AI Agent fails, it can maintain system functionality through task reallocation. Multi-AI Agents are based on modular design rules, and all AI Agents' decisions and task execution can be traced and interpretable. In terms of scalability, multi-AI Agent systems can automatically adjust their behavior and strategies based on changes in the environment and task requirements, adapting to the complexity of tasks by increasing or decreasing the number of AI Agents, and complementing and expanding application scenarios.

3 Multi-AI Agent Collaboration

3.1 Theoretical basis

3.1.1 Collaborative mechanism. Multi-AI Agent collaborative system consists of multiple autonomous, interactive, and collaborative AI Agents that achieve collaboration through dynamic task decomposition, distributed resource scheduling, and reasonable task allocation capabilities. The agent layer is composed of multiple independent AI Agents, each with unique professional knowledge and resources, and capable of performing different types of tasks separately. At the model layer, lightweight models are responsible for fast response tasks such as environment perception and semantic understanding. The LLM model focuses on planning the task planning, allocation, and other processes of AI Agents. Multi modal model assisted tool agents are responsible for task execution. Collaboration layer, multi-AI Agent collaboration types include cooperative, competitive, and hybrid [11]. The communication between AI Agents covers centralized, distributed, and hierarchical structures; The multi-AI Agent collaboration strategy includes

rule-based, role-based, and model-based strategies. The specific collaborative mechanism is shown in Figure 2.

3.1.2 Multi-AI Agent architecture design. Multi-AI Agent architecture includes multi-AI Agent horizontal architecture and multi-AI Agent vertical architecture. In multi-AI Agent vertical architecture, one AI Agent serves as the leader and other AI Agents report directly to it. In multi-AI Agent horizontal architecture, all AI Agents participate equally and discuss tasks together. Multi-AI Agent frameworks continue to emerge, and typical multi-AI Agent system architectures [12] include point-to-point architecture, centralized architecture, distributed architecture, hybrid architecture, etc.

The research on multi-AI Agent frameworks shows a clear upward trend, and the different interaction and collaboration modes between multi-AI Agent systems make the processing of complex tasks more efficient and simple. MetaGPT [13], as an open-source multi-AI Agent framework, role assigns standard operating procedure(SOP) and encodes it into the AI Agent architecture, significantly reducing the difficulty of solving complex tasks. The core of AutoGen [14] consists of two types: User-Agent and Assistant-Agent, dedicated to researching solutions in the field of software development. To address the complex challenges faced by developers in multi-AI Agent system orchestration, OpenAI has launched the open-source multi-AI Agent framework Swarm, which greatly simplifies the coordination and management of multiple AI agents, enabling more effective deployment, orchestration, and testing of agent based systems. The agentUniverse framework provides multi-AI Agent collaboration orchestration components, allowing developers to customize multi-AI Agent collaboration modes and achieve multi-step disassembly and step-by-step execution of complex problems. The universal AI Agent product Manus is based on a multi-AI Agent collaborative architecture and has full chain autonomous execution capability. The emergence of multi-AI Agent frameworks such as AgentVerse [15], CrewAI, LangGraph, Magentic One, PEER [16] demonstrates the rapid development and diverse applications of multi-AI Agent systems.

3.1.3 Communication protocol. In the process of multi-AI Agent collaboration, multiple AI Agents communicate with each other through protocols. However, in the process of multi-AI Agent communication, there are also challenges such as single deployment function, low collaboration efficiency, formation of data islands, and complex development processes. To solve these problems, communication protocol design solutions have emerged. One is interconnectivity, where all AI Agents should be able to communicate with each other and break down data silos. The second is the native interface, AI should interact with the digital world in its preferred way, such as APIs or communication protocols. The third is efficient collaboration, where AI Agents should achieve self-organization and self negotiation to build a lower cost and more efficient collaborative network. Marro et al. [17] proposed a meta protocol called Agora, which effectively balances multifunctionality, efficiency, and portability in multi-AI Agent communication.

Agent network protocol (ANP) and model context protocol (MCP) [18] have important research value and application prospects in the field of AI Agent communication. ANP is an open protocol framework designed for AI Agent-based Web, with AI Agents as the core and more emphasis on equal collaboration among AI Agents,

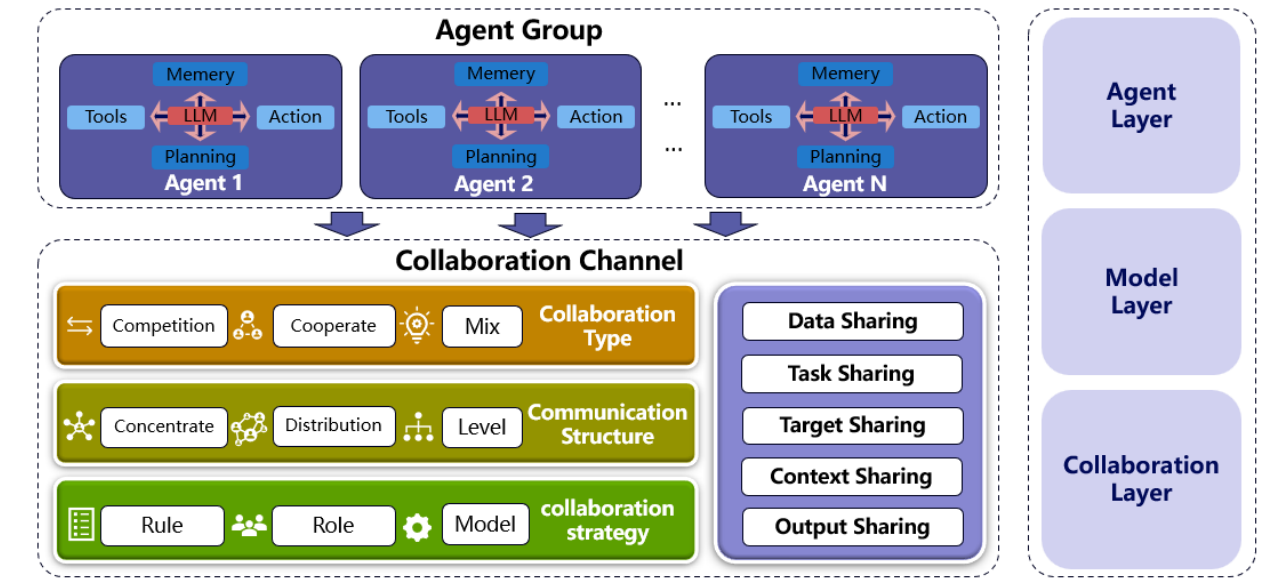


Figure 2: Multi-AI Agent collaborative mechanism.

aiming to build a decentralized agent collaboration network. ANP implements decentralized authentication and allows any two AI Agents to connect. MCP takes the model as its core and simplifies the integration process of algorithm models, platform tools, data resources, etc. By providing a common interface, it significantly improves the communication efficiency, security and stability between LLMs and external objects. Researchers can choose the right type of communication protocol according to specific application scenarios and requirements.

3.2 Key technologies

3.2.1 Collaborative technology. The core of multi-AI Agent collaboration is how to design effective mechanisms that allow AI Agents to coordinate their actions effectively. In terms of task allocation, it includes game theory-based task allocation method, optimization-based task allocation and market-based task allocation. In terms of resource allocation, model-based dynamic resource allocation constructs a network environment model and a resource allocation model, and the AI Agent optimizes its resource allocation decision based on state information inputs. Rule-based resource allocation allocates resources to agents based on pre-defined rules and policies. For example, resources are allocated based on the priority and task requirements of the AI Agent. In learning-based resource allocation, the AI Agent dynamically adjusts its resource allocation strategy based on historical experience and environmental feedback through a learning algorithm to adapt to complex environmental changes. Qin et al. [19] modeled multi-AI Agent task allocation as a game problem, where each AI Agent selects tasks based on utility functions and dynamically adjusts and optimizes them using Nash Equilibrium theory. Kai et al. [20] constructed an optimization model with task completion efficiency and resource

consumption as optimization objectives, and allocated tasks by solving optimization problems.

3.2.2 Learning technology. Multi agent reinforcement learning (MARL) is an extension of reinforcement learning in multi-AI Agent systems that aims to learn optimal decision strategies through the interaction between multiple agents and the environment. It is suitable for solving various complex multi-AI Agent sequential decision problems [21]. According to different research directions, multi-AI Agent reinforcement learning can be categorized into value function-based and policy-based approaches. It can also be categorized into cooperative, competitive and hybrid task types based on the task type [22]. In addition, it can also be categorized based on whether a communication mechanism is established or not.

MARL has received a lot of attention in recent years. Ma et al. [23] proposed an efficient reinforcement learning method for decentralized collaborative training and decision making, which reduces the dependence on global communication and large amounts of interactive data, and is expected to improve the decision-making performance of large-scale systems. Lowe et al. [24] proposed a centralized training and decentralized execution multi agent deep determining policy gradient (MADDPG) algorithm, which has significant advantages in multi-AI Agent interaction. With the development of deep learning technology, multi agent deep reinforcement learning (MADRL) has made significant progress in multiple fields. Foerster et al. [25] designed counterfactual multi-agent policy gradient methods to solve the cooperation and competition problems in multi-AI Agent environments. Huang et al. [26] proposed a multi-AI Agent reinforcement learning method based on deep spatiotemporal attention neural network to determine traffic signal timing in large-scale transportation networks.

Table 1: Summary of AI Agent industry and application scenarios.

Industry	Scenario Application		Related Research
Manufacturing	Process Control	Equipment Monitoring	TARS-RPA-Agent
	Logistics Scheduling	Production Optimization	Resource Agent (RA) [33]
	Quality Inspection	Fault Diagnosis	...
	Equipment maintenance	...	...
Medical	Intelligent Guidance	Specialized Surgery	Healthy Care Agents
	Proposal Proposal	Health monitoring	DrugAgent [34]
	Risk Assessment	Drugs Inquiries	MDAgents [35]
	Trend Analysis	...	...
Transportation	Autonomous Driving	Flow Monitoring	MATSim [36]
	Intelligent Dispatching	Emergency Response	TP-GPT [37]
	Parking Control	Traffic Control	CACC [38]
	Road Administration	...	...
Finance	Investment Advisory [39]	Marketing Services	BondGPT+
	Claims Expert	Insurance Advisor	MADDQN [40]
	Financial Editor	Customer Service	FinRobot [41]
	Financial Reports	...	...
...	...	...	...

3.2.3 Security technology. As the frequency and scale of cyber-attacks increase, multi-AI Agent systems are exposed to a variety of security threats such as malicious attacks, information leaks and system failures. Researchers are committed to focusing on the security mechanisms and defense technologies of multi-AI Agent systems. Du et al. [27] proposed a novel framework, scalable secure multi agent reinforcement learning (SS-MARL), aimed at improving the security and scalability of MARL methods. Shi et al. [28] proposed a distributed iterative localization algorithm based on detection and retention strategy, which can effectively resist false data injection attacks and ensure the stable operation of sensor networks. To solve the problem of obvious rejection intentions in large language model replies that are easily recognized by attack models, Xu et al. [29] proposed a multi-AI Agent attacker camouflage game method, which enables the model to securely reply to attackers while hiding defense intentions, enhancing the model's ability to camouflage defense intentions to adapt to dynamic attack and defense environments.

3.3 Scenario application

Multi-AI Agent collaboration technology has successfully compensated for the shortcomings of single AI Agent in dealing with complex scenarios, providing new directions and solutions for technology applications in related fields. AI Agents have deeply penetrated into multiple industries such as basic science, finance, industry, education, and healthcare, empowering various scenarios such as research and development design, customer service, production and manufacturing [30], and business management, greatly improving work efficiency and social productivity. OpenAI has released an intelligent operator that can operate computers, write code, book travel, automate e-commerce shopping, and perform multiple tasks. Colabianchi et al. [31] are dedicated to the research of conversational chatbots in the manufacturing industry, accelerating the process of digital to intelligent transformation. IBM Watson agent

can help doctors develop personalized cancer treatment plans. Autonomous vehicles [32] can share real-time information such as speed and location through multi-AI Agent collaboration, optimize traffic flow, and reduce congestion and accidents. Table 1 shows the application performance of multi-AI Agent systems in various industries.

4 Research Findings

4.1 Development advantages and disadvantages

4.1.1 Advantages. The fundamental theoretical research on multi-AI Agent collaboration lays a solid foundation for technological applications. Multi agent collaboration can effectively overcome the limitations of a single AI Agent. Multi-AI Agent systems specialize LLMs into various AI Agents with different abilities, enabling these diverse AI Agents to communicate and collaborate with each other to complete a task, effectively simulating complex real-world environments. The autonomy, reactivity, and proactivity of agents give them unique advantages in scientific research, and their research value is mainly reflected in theoretical exploration, human-computer interaction, collaborative optimization, and other aspects. At the same time, evaluations on multi-AI Agent task sets such as GAIA [42], TAU Bench [43], SWE Bench [44] are constantly emerging, providing quantitative standards for subsequent engineering optimization.

Multi-AI Agent collaboration technology, learning technology, and security technology provide strong support for the development of AI Agent. Multi-AI Agent can handle more complex problems through division of labor and collaboration, significantly improving the robustness, flexibility, and adaptability of the system, while achieving dual improvements in task processing quality and efficiency. Multi-AI Agent collaboration can flexibly configure tasks according to different security requirements, ensuring that agents achieve a balance between security and optimality.

Multi-AI Agent collaboration solves the collaborative needs of complex scenarios. Multi-AI Agent has significant advantages in handling complex and dynamic tasks, providing effective solutions for problems in environments that require high coordination and adaptability, such as smart grids, smart cities, industrial manufacturing, traffic management, supply chains, and other fields. Meanwhile, through distributed computing power scheduling and dynamic load balancing, multi-AI Agent systems can reduce inference costs and improve energy efficiency, optimize resource utilization, and align with the trend of green computing.

4.1.2 Disadvantages. With the further popularization of LLM technology and AI Agent products, while receiving high attention from various parties such as industry, academia, research and application, multi-AI Agent collaboration still faces multidimensional constraints in technology, application, security and other aspects.

Heterogeneous coordination and autonomy face challenges. With the expansion of system scale, the heterogeneity of multi-AI Agent systems increases, making it difficult for agents to work efficiently together. The autonomy of AI Agent makes it difficult to be fully controlled in open environments, and excessive autonomy may lead to agents deviating from predetermined goals or behavioral patterns.

Communication and Information sharing barriers. Multi-AI Agent systems require efficient information sharing mechanisms, but communication delays, packet loss and bandwidth constraints can affect the transfer of information between agents.

Potential problems with adaptability and personalization. Multi-AI Agent systems need to adapt to a variety of complex application scenarios and meet the personalized needs of users to provide better services and experiences. The adoption threshold of agent applications is high and there is an urgent need to strengthen the application ecosystem and platform tools.

Trust mechanisms and global security are key challenges. When AI Agents operate in complex conditions or unknown environments, they are vulnerable to malicious exploitation and may lead to unfavorable outcomes. It is difficult for agents to avoid issues such as privacy when interacting with humans and accessing users' personal information.

4.2 Innovation direction

Efficient collaboration among multiple agents, optimization of AI Agent workflows, efficient human-machine collaboration, and multimodal fusion have become important development directions. Agentic AI, represented by agent technology, heralds a giant leap in upgrading AI services from assistance to autonomy. Multi-AI Agent collaboration technology will be deeply integrated with human activities, and efficient human-agent collaboration will be realized through the design of reasonable collaboration mechanisms.

The AI Agent application ecosystem and platform tools must be continuously strengthened, and the deployment threshold of multi-AI Agent applications must be continuously lowered. By improving the flexibility and adaptability of AI Agents, the breadth and depth of application scenarios will be further expanded to meet personalized needs and provide users with a better subjective experience.

Improving the privacy information and security protection mechanism to ensure the credibility and controllability of the technology will become an important guarantee for the development of AI Agents. Incorporating fairness and ethical considerations into the decision-making process of multi-AI Agents will ensure that the system's decisions are not only efficient, but also ethical and moral.

Last but not least, we should strengthen the cross-border integration of industries, enhance interdisciplinary research and cooperation and communication among industry, academia, research and application parties, break through data bottlenecks and product form limitations, realize large-scale applications, and open up a wide application space in various industries. Accelerate the construction of an agent application support system, such as promoting the development and implementation of AI Agent standards, strengthening the construction of open source ecosystems, and establishing evaluation and testing systems.

5 Conclusion

As LLMs and AI Agent technologies continue to be developed and implemented, multi-AI Agent collaboration has become a focus of current research. This paper comprehensively reviews multi-AI Agent collaboration and outlines the research foundations, such as the core capabilities of AI Agent. It analyses the basic theories of multi-AI Agent collaboration mechanism, architecture design and communication protocols, summarizes in-depth multi-AI Agent collaboration technology, learning technology and security technology, comprehensively grasps the application scenarios of agent collaboration, discusses the advantages and disadvantages of the development of multi-AI Agent collaboration, puts forward the innovative research direction, and looks forward to the future research frontiers.

Multi-AI Agent collaboration is an area that combines cutting-edge academic research with practical engineering value. In the future, with the increasing maturity of multi-AI Agent collaboration technology, the boundaries of multi-AI Agent technology and applications will continue to expand, taking a more solid step on the road to solving complex human problems, deeply empowering multi-domain segmented task scenarios, and comprehensively empowering the development of new high-quality productivity.

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